# COMPLETED LOCAL RINGS OF A PUSHOUT ALONG A FINITE CLOSED SUBSCHEME

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*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-curves/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* The counted proof is the author's complete argument: the Algebra completion lemmas are external tooling (stubs); the starting diagram comes from the situation paragraph directly above the lemma (shared section setup, same convention as the situation-ABC cards).

*Editorial note.* The label lemma-complete-local-ring-pushout names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

*Editorial note.* The counted statement is carried by the author's own lemma environment in the chapter source.

*Allowed standard prerequisites (editorial).*

- Pushout construction. Same unit, situation paragraph preceding the lemma. The proof starts from the ring-level fibre product diagram of the pushout construction ("by construction").
- Completion tooling. Algebra, Lemma algebra-lemma-completion-flat and Lemma algebra-lemma-completion-finite-extension (standard tooling). Completion preserves the fibre product diagram for finite algebras; the finite-extension completion lemma identifies the completed rings of A'.

*Source reference.* Original lemma 07N9: sources/stacks-chow/tags.txt, source lines 9754–9754. Prerequisite pointer invoked at proof line(s) 3167; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

*Source reference.* Original lemma 00MB: sources/stacks-chow/tags.txt, source lines 817–817. Prerequisite pointer invoked at proof line(s) 3165; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

**Lemma 0.1.** *In the situation above, let* $Z = \mathrm{Spec}(k')$ *where* $k'$ *is a field and* $Z' = \mathrm{Spec}(k'_1 \times \ldots \times k'_n)$ *with* $k'_i/k'$ *finite extensions of fields. Let* $x \in X$ *be the image of* $Z \to X$ *and* $x'_i \in X'$ *the image of* $\mathrm{Spec}(k'_i) \to X'$*. Then we have a fibre product diagram*

$$\begin{array}{ccc} \prod_{i=1,\ldots,n} k'_i & \longleftarrow & \prod_{i=1,\ldots,n} \mathcal{O}^\wedge_{X',x'_i} \\ \uparrow & & \uparrow \\ k' & \longleftarrow & \mathcal{O}^\wedge_{X,x} \end{array}$$

*where the horizontal arrows are given by the maps to the residue fields.*

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*Proof.* Choose an affine open neighbourhood $\mathrm{Spec}(A)$ of $x$ in $X$. Let $\mathrm{Spec}(A') \subset X'$ be the inverse image. By construction we have a fibre product diagram

$$\begin{array}{ccc} \prod_{i=1,\ldots,n} k_i' & \longleftarrow & A' \\ \uparrow & & \uparrow \\ k' & \longleftarrow & A \end{array}$$

Since everything is finite over $A$ we see that the diagram remains a fibre product diagram after completion with respect to the maximal ideal $\mathfrak{m} \subset A$ corresponding to $x$ (Algebra, Lemma 00MB). Finally, apply Algebra, Lemma 07N9 to identify the completion of $A'$. $\square$